

Brain-Computer Interfaces

ISSN: 2326-263X (Print) 2326-2621 (Online) Journal homepage: www.tandfonline.com/journals/tbci20

Coherence-based connectivity analysis of EEG and EMG signals during reach-to-grasp movement involving two weights

Cristian D. Guerrero-Mendez & Andres F. Ruiz-Olaya

To cite this article: Cristian D. Guerrero-Mendez & Andres F. Ruiz-Olaya (2022) Coherence-based connectivity analysis of EEG and EMG signals during reach-to-grasp movement involving two weights, Brain-Computer Interfaces, 9:3, 140-154, DOI: 10.1080/2326263X.2022.2029308

To link to this article: https://doi.org/10.1080/2326263X.2022.2029308

© 2022 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

Published online: 12 Feb 2022.

Submit your article to this journal

Article views: 4321

View related articles

View Crossmark data

Citing articles: 20 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tbci21

BRAIN-COMPUTER INTERFACES
2022, VOL. 9, NO. 3, 140–154
https://doi.org/10.1080/2326263X.2022.2029308

Taylor & Francis
Taylor & Francis Group

ORIGINAL RESEARCH

OPEN ACCESS Check for updates

Coherence-based connectivity analysis of EEG and EMG signals during reach-to-grasp movement involving two weights

Cristian D. Guerrero-Mendez and Andres F. Ruiz-Olaya

Bioengineering Research Group, Faculty of Mechanical, Electronic and Biomedical Engineering, Antonio Nariño University, Bogotá, Colombia

ABSTRACT

Corticomuscular coherence allows studying the mechanism of the cerebral cortex's control of muscle activity, which reveals the communication in corticospinal pathways between the primary motor cortex and muscles. The present study aims to quantify the connectivity between the motor cortex (EEG signals) and five muscles of the right upper limb (EMG signals) during the manipulation of an object. A public dataset (WAY-EEG-GAL) was used which recorded EEG and EMG of twelve healthy subjects who performed movements of reaching, grasping, holding, and replacing objects of two different weights. Corticomuscular connectivity was established using the coherence algorithm for 3 EEG channels and 5 upper-limb muscles varying two objects' weights (0.165 kg and 0.660 kg). Results show that the 0.165 kg weight shows greater coherence between the signals for all analyses than the 0.660 kg weight. Furthermore, the results show that there is a contralateral and ipsilateral behavior in the EEG-EMG coherence.

ARTICLE HISTORY
Received 17 August 2021
Accepted 7 January 2022

KEYWORDS
Corticomuscular coherence; object manipulation; functional connectivity; hybrid brain-computer interface; motor execution; electroencephalography (EEG); electromyography (EMG); reach-to-grasp

1. Introduction

Previous reports have shown that synchronization between neurons in the motor cortex and motor units occurs during the performance of a motor task [1]. That mechanism was shown using one Magnetoencephalography (MEG) channel recording the cortical motor activity and the surface electromyogram (EMG) of a contralateral active muscle during the execution of a muscular voluntary contraction; Corticomuscular connectivity between cortical rhythms and rectified EMG confined to the beta (15–30 Hz) frequency range was evidenced, applying coherence analysis [2]. Furthermore, movements have long been known to induce frequency-specific changes in Electroencephalography (EEG) [3]. Those works evidence that there is corticomuscular connectivity defined as a relationship, association, or statistical dependence between EEG and EMG signals resulting from the functional integration of the neural and muscular systems [4,5]. EEG–EMG coherence could be used to examine a functional connection between a human brain and muscles by calculating the linear relationship of frequency domain components of EEG and EMG signals. Corticomuscular coherence allows studying of the mechanism of the cerebral cortex's control of muscle activity, which reveals the communication in corticospinal pathways between the primary motor cortex and muscles. Normally, cortical events propagate to the periphery, and the motor cortex also receives input from the periphery [6].

Corticomuscular coherence is task-dependent, that is reflects attention and precision, compliance of the gripped objects, displacement, magnitude of the force, and learning processes [2]. In the literature, it has been reported that cortico-muscular coherence is higher in beta (15–30 Hz) and gamma (30–80 Hz) frequency bands of EEG signals [7,8]. Furthermore, coherence of the beta band increases during postural tasks by maintaining sustained motor contractions [6]. Corticomuscular coherence has also been observed at higher gamma band frequencies during dynamic movements or during increasing muscle contraction strength [3].

Other studies have demonstrated that grasp is encoded by neural activity [9]. Kim *et al.* describe the association between EEG and EMG signals in healthy individuals when subjects performed active exercise (finger motion) with movement intention and passive exercise without movement intention. These findings show how movement intention in the patients enhances association EEG–EMG, to be implemented in a rehabilitative training system [10,11]. Various works have quantified connectivity in tasks related to holding positions and manipulating objects by applying different forces [4,12,13]. Nevertheless, brain connectivity

CONTACT Cristian D. Guerrero-Mendez crguerrero69@uan.edu.co Bioengineering Research Group, Faculty of Mechanical, Electronic and Biomedical Engineering, Antonio Nariño University, Bogotá, Colombia.

© 2022 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

studies have been reported to assess selective attention [14], and others have used connectivity to identify movements, where improvements for classification rates have been found with connectivity-based methods over standard methods [15,16].

Neuroprosthetics using neural information from the patient using EMG and EEG, could be envisaged to enhance control of rehabilitation devices and improve usability. Thus, the EMG signal from a voluntary muscle contraction allows the user to transmit the intention of a command, which is related to EEG neuronal activity to perform a motor action. Likewise, with the use of multimodal information, conventional BCI systems could improve their performance in terms of classification rate or accuracy, using what is currently called hybrid Brain–Computer Interfaces (hBCI) [17,18].

The restoration of functional manipulative activity is of special importance in the rehabilitation of people with amputation or motor disability at the level of the upper limb, taking into account their involvement in the performance of activities of daily living. The manipulation of an object requires executing a series of motor tasks that include reaching, grasping, holding, and replacing [19]. Such tasks require the coordinated and timed activation of the muscles that intervene in them. Identifying and predicting motor tasks during reach and grasp for manipulating an object increases the usability, comfort, and controllability of the prosthetic device [20]. This translates into greater user acceptance in rehabilitation systems based on Brain–Computer Interface (BCI) systems, so it is of great importance to identify and predict these events. An adequate identification of the different stages during the manipulation of an object would allow a lower latency in the response of BCI systems, compared to the identification that is performed when the upper limb is already in the final posture.

To our knowledge, currently, it is not clear how the EEG–EMG corticomuscular coherence is related to reach-grasp-lift-hold and replace tasks during the manipulation of an object. This work focuses on quantifying the coherence EEG–EMG, under several conditions that include: 1) Evaluating EEG-bands of extended alpha (α) $(6-13)$ Hz, beta (β) $(14-30)$ Hz and gamma (γ) $(35-50)$ Hz that contributes to coherence; 2) Evaluating five upper-limb muscles during manipulation of an object involving two weights. A public dataset was used that records EEG and EMG information during the performance of tasks of reaching and grasping objects of different weights. The analysis was defined at two weights: 0.165 kg and 0.660 kg, to evaluate the effect of manipulating an object that requires the generation of a low and a high muscle contraction.

First of all, in this study, in order to characterize the cortical response, Event-Related Desynchronization (ERD) and Event-Related Synchronization (ERS) were used. ERD: Increased cellular excitability in thalamocortical systems results in a low-amplitude desynchronized EEG [21,22]. In normal participants, ERD and ERS are considered to indicate the activation and subsequent recovery of the motor cortex during planning, executing, and completing a movement. Therefore, the amount of the cortical activation during a sensory, motor or cognitive task could be indexed by the ERD and ERS concerning a baseline [23]. To characterize the muscular electrical activity, the EMG signals were processed using segmentation and amplitude estimation techniques by calculating the Root Mean Square (RMS) metric. After characterizing the cortical responses and muscle electrical activation, the estimation of coherence was performed by evaluating different studies mentioned above, where the frequency bands with greater coherence were identified and the muscles that present greater coherence with the EEG channels.

This article is organized as follows: Section II presents the experimental methods, including the experimental protocol, the description of the data processing, and the algorithm proposal; Section III presents the results obtained; and finally, Section IV presents the discussion that includes the conclusions, impact of the results, and future works.

2. Materials and methods

2.1. EEG–EMG dataset

This work was implemented using the WAY-EEG-GAL dataset, which is an open and free available EEG–EMG dataset [24]. The dataset consists of EEG and EMG recordings, as well as 3D hand and object position measurements. Twelve healthy right-handed subjects (8 females and 4 males, aged 19–35 years) were recorded using 32 EEG channels located at *Fp*1, *Fp*2, *F*7, *F*3, *Fz*, *F*4, *F*8, *FC*5, *FC*1, *FC*2, *FC*6, *T*7, C_3, C_z, C_4, *T*8, *TP*9, *CP*5, *CP*1, *CP*2, *CP*6, *TP*10, *P*7, *P*3, *Pz*, *P*4, *P*8, *PO*9, *O*1, *Oz*, *O*2, *PO*10 according to the 10–20 international EEG placement system. Reference and ground electrodes were connected to *FCz* and *AFz* locations, respectively. In addition, five EMG channels from the following muscles: 1. Anterior Deltoid (AD), 2. Brachioradialis (B), 3. Flexor Figitorum (FD), 4. Common Extensor Digitorum (CED), and 5. First Dorsal Interosseous

(FDI). EEG signals were recorded with the ActiCap device at a sampling rate of 500 Hz. On the other hand, EMG signals were recorded using five sensors at a sampling frequency of 4 kHz. Figure 1 shows the experimental setup for the dataset acquisition.

In the protocol, initially, there is a rest period of 2 seconds before starting the movement where subjects maintain the right upper limb leaning on a table, next, the subject receives a visual indication from a LED to start performing a reaching movement of the right hand toward an object. Then, the user grasps it with the index and thumb fingers; afterward lifts it and holds the object steadily within a circle that is about 5 cm from the table for 2 seconds until the LED turned off, and subsequently replacing the object and returns the upper limb to the position indicator, as shown in Figure 1. The object varied in weight and contact surface randomly in 3 different conditions, the variation of weights was kept at (0.165, 0.330, 0.660 kg) and contact surface at (sandpaper, chamois, silk). To module the objects' weight, an electromagnet was used to present the same weight between 1 and 4 times and then change it. These weight variations were performed using two electromagnets of 0.165 kg and 0.330 kg that to vary the weight in some conditions only one electromagnet is activated and in others both. For the variation of the contact surface, the same logic was maintained, but in this condition, an external person had to intervene on the object. Finally, ten series of approximately 32 trials were recorded, for a total of 328 trials per participant in which the weight of the object (0.165, 0.330, 0.660 kg), the contact surface (sandpaper, chamois, silk), or both was changed.

This study used 3 EEG channels (C_3, C_z, and C_4) and five EMG channels for calculating corticomuscular connectivity, following the international 10–20 system of EEG electrode placement and muscle location for EMG, as presented in Figure 2. Data from two different weights were used when the subject manipulated the object (0.165 and 0.660 kg) where the contact surface was kept constant with sandpaper. Five series of weights were used where each series included 22 trials, for a total of 110 trials per participant. Finally, data were taken for each trial until the subject performed the task of replacing the object.

2.2. Dataset integrity validation

Data integrity validation of EEG signals was performed to verify contralateral behavior and characterize cortical responses when performing the movements with the right upper limb. For this purpose, the EEG signals were filtered using an 8*th* order bandpass Butterworth filter between $8-30$ Hz, and a Common Average Reference Filter (CAR) to remove related noise on the electrodes located in the cortico-motor area. Subsequently, the EEG signals' Power Spectral Density (PSD) is calculated for all 32 channels (Figure 2) recorded in the two weights to evidence the frequency distribution across the cortico-motor area. For this, the PSD was calculated using the Fast Fourier Transform (FFT) with a Hanning window of 1 second in the frequency range $8-30$ Hz for the execution data (2–8 seconds).

In addition, Event-Related Synchronization and Desynchronization (ERS/ERD) was calculated for 3 EEG channels (C_3, C_z, and C_4) to identify the short-lasting and localized amplitude attenuation of EEG rhythms within the α and β bands [23]. EEG signals were processed for each weight using the open-source FieldTrip Toolbox (https://www.fieldtriptoolbox.org/). The ERS/ERD was implemented with a time-sliding window in time steps of 50 ms with frequency intervals of 1 Hz from 8 to 30 Hz using the Morlet Wavelet method for a time-frequency representation. The ERD is related to a decrease in amplitude and power and the ERS is related to an increase in amplitude and power, which was calculated using equation (1) with a baseline period of 1 second before the stimulus [23]. ERS/ERD

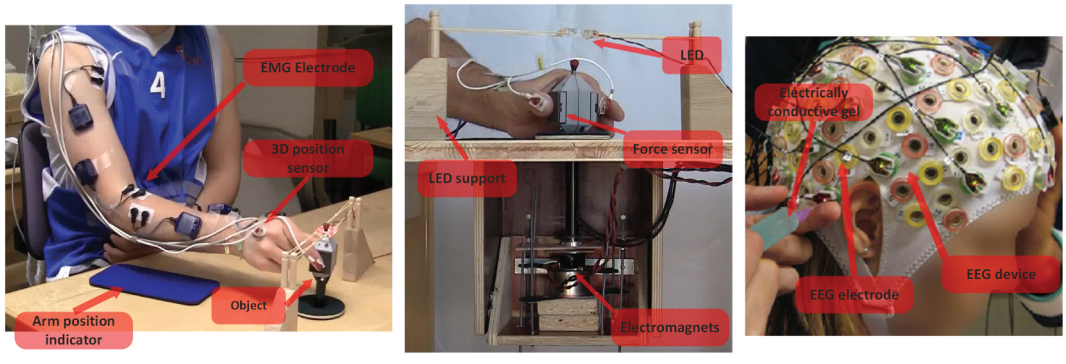


Figure 1. Experimental setup for data acquisition. Modified image and adapted from [24] available under the terms of Attribution 4.0 International Creative Commons License (https://creativecommons.org/licenses/by/4.0/).

data is presented as a percentage of the baseline period. Nevertheless, only the significant data of the increase or decrease power (ERS/ERD) with respect to a reference time of 1 second were taken for the analysis. For this, the Bootstrap algorithm was used with a significance level of 0.05 ($p < 0.05$) according to the description made by [25]. Additionally, to evaluate contralateral behavior in the execution of the movement, a statistical analysis was performed using the ERS/ERD data to identify significant differences between channels and weights using the Mann–Whitney U-test with $p - value$ of 0.05, after checking for abnormal distribution of the data.

$$ERD\% = \frac{R - A}{A} \times 100 \tag{1}$$

In equation (1), A is the power related to the step time after the event between the interest frequency range and R is the reference baseline period.

Data integrity validation of EMG signals was performed to verify the amplitude and characterize the muscle electrical activity when the movement was performed. For this, EMG signals for the five muscles (Figure 2) were filtered using an $8th$ order bandpass Butterworth filter between $10 - 500$ Hz due to in that frequency band there is higher energy and power of the EMG signal [26]. Subsequently, EMG signals were analyzed through the signal amplitude estimation using the RMS value following the equation (2) with a sliding window of 300 ms overlapped 50%. In the analysis, when the subject starts to perform the movement, the EMG amplitude should be greater for the 0.660 kg weight than for the 0.165 kg weight, where the amplitude of the signal estimates the muscle activation when a task is performed [27]. For this, a statistical analysis was applied using the Mann–Whitney U-test with $p = 0.05$ using the mean of all subjects, due to the muscle activity data of the five muscles and the two weights do not present a normal distribution according to the Kolmogorov–Smirnov test. In equation (2), N is the length of the window, and X_n is the EMG signal for each sample n.

$$RMS = \sqrt{\frac{1}{N}\sum_{n=1}^{N} X_n^2} \tag{2}$$

2.3. Methodology for EEG–EMG coherence analysis

After dataset integrity validation the methodology for EEG–EMG coherence analysis presented in Figure (3) was implemented. First, the study is delimited using two weights and selecting subjects. After, artifacts are detected in the EEG signal, and channels are selected. Next, the signals are pre-processed using filters. After, the EEG and EMG signals are segmented by using a Hanning window of 1 second overlapped to 25% for calculating the coherence, which is represented using significant coherence values. And finally, statistical

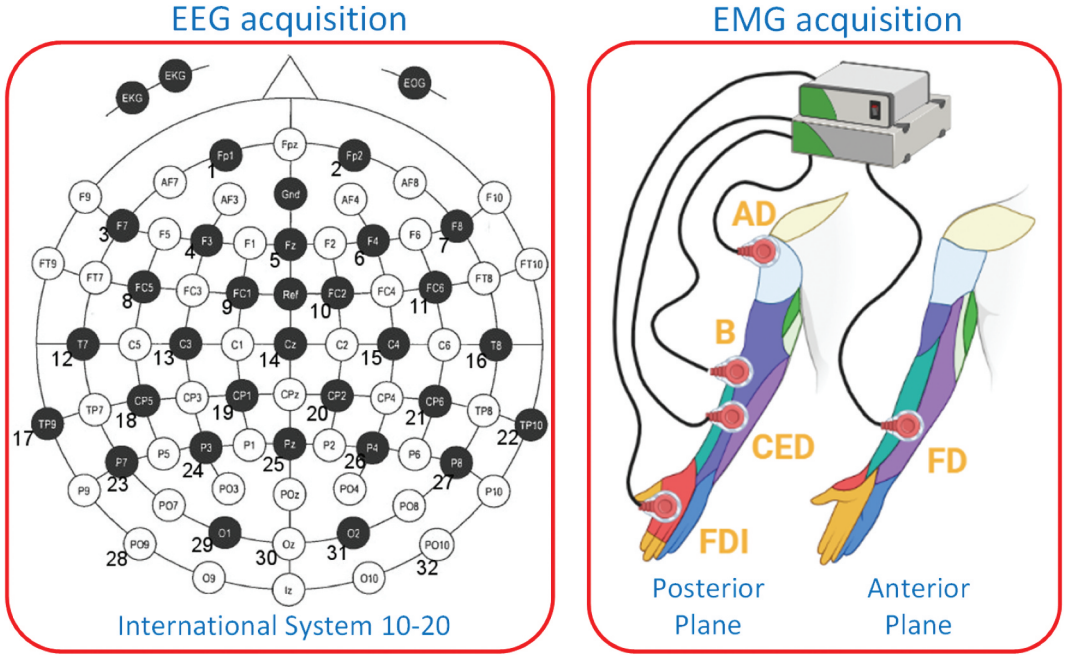


Figure 2. EEG electrodes layout following the international 10–20 system used in the original WAY-EEG-GAL dataset and EMG electrode placement following the locations of five upper limb muscles.

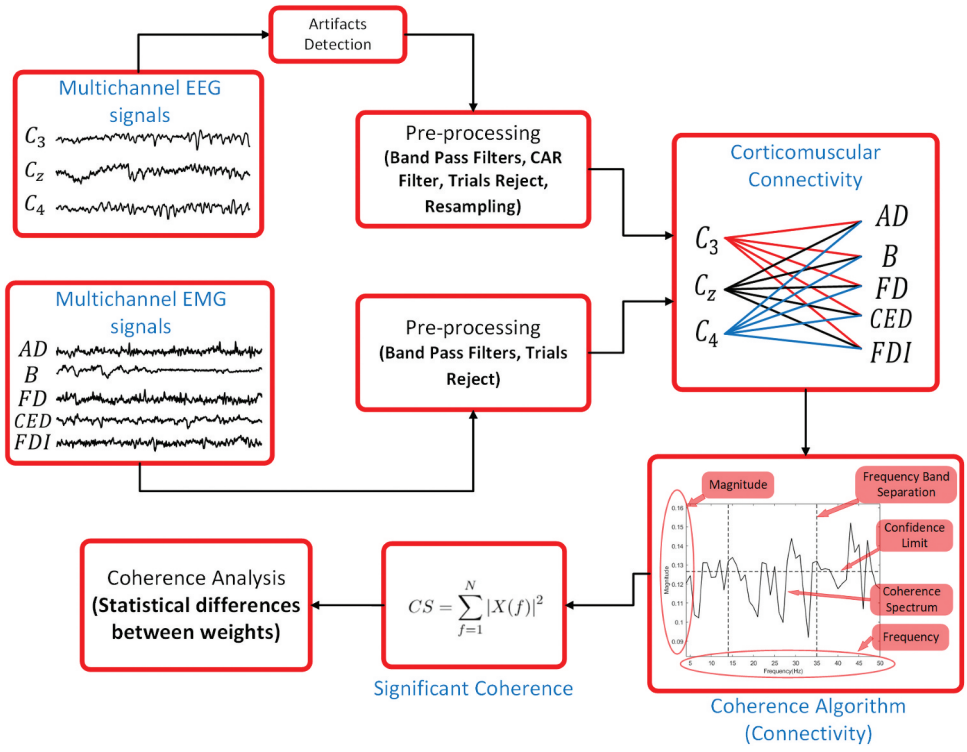


Figure 3. Block diagram of the implemented methodology in this study to estimate corticomuscular connectivity between EEG and EMG signals during the movement of reach-grasp-lift-hold and replace an object.

analysis is performed to find significant differences between the coherence results for the two weights using the paired t-test.

2.3.1. Subjects and channels selection

In this study, a trial rejection process was performed by identifying artifacts in EEG signals, using threshold criteria. For this, the EEG signals should be within the range of $\pm$ 350μV [28]; otherwise, trials were rejected containing high outliers that were outside the range. If outliers are detected in an EEG channel, the trial is rejected for all EEG and EMG channels. EEG channels C_3, C_z, and C_4 and all EMG channels were analyzed. The subject selection criterion was that amount rejects trials no should be greater than 10% of the total trials [29]. In this study, no subject was rejected, selecting all participants.

2.3.2. Pre-Processing

To estimate corticomuscular connectivity, the EEG signals for each trial and each subject were filtered using a bandpass Butterworth filter of 8*th* order between 6 – 50 Hz and a CAR filter. The EMG signals were filtered using a bandpass Butterworth filter of 8*th* order the 6 – 50 Hz. Contaminated trials in both EEG and EMG signals were rejected for each subject. A resampling of EEG signal was performed at the sample rate of EMG signal that is 4 kHz, such as reported in the literature [30]. For this purpose, only the execution movement is analyzed, because at rest there is no movement intention and therefore muscle activation is negligible [4]. Some authors describe that coherence in the resting is insignificant [31].

2.3.3. Connectivity analysis between EEG and EMG signals

Connectivity analysis between EEG and EMG signals was calculated using the coherence algorithm, a measure of connection or correlation between two signals in the frequency domain that determines the strength of correlation in the range of 0–1 [6]. Coherence was implemented to evaluate the connectivity between the electrical activity of the brain and muscles when performing an object manipulation movement. To calculate coherence, the frequency components of EEG and EMG signals in the range 6 – 50 Hz were extracted by calculating the auto-spectrum and cross-spectrum using MATLAB (Version R2020b, MathWorks Inc). EEG and EMG signals were segmented by a 1 second Hanning window with 25% overlap at a frequency resolution of 2 Hz between 6 and 50 Hz for each trial and subject [32]. Combinations between channels were obtained by relating the 3 EEG channels (C_3, C_4, C_z) with each EMG

channel (5 Channels), which generate a total of 15 combinations between channels, as shown in Figure 3. The coherence was calculated according to equation (3) [33].

$$Coh_{xy}(f) = \frac{\left|P_{xy}(f)\right|^2}{P_{xx}(f) \cdot P_{yy}(f)} \tag{3}$$

where $Coh_{xy}(f)$ is the correlation specified in the range 0–1 between two signals in the frequencies f. $P_{xy}(f)$ is the cross-spectrum of x and y, while $P_{xx}(f)$ and $P_{yy}(f)$ is the auto-spectrum of x and y, respectively, which uses Welch's averaged and modified periodogram method of spectral estimation. In addition, a significant threshold was calculated using the coherence values to use in the study the significant results of the proposed coherence method, equation (4) [33].

$$Confidence\ Limit = 1 - (1 - \varepsilon)^{\frac{1}{L-1}} \tag{4}$$

where L is the number of segments and ε is the confidence level. However, in this study, L is the number of windows for each weight, and the confidence level used is 95% for both weights.

To extract significant coherence, a conditional-based classification was implemented where the coherence spectrum above the confidence limit was analyzed. Then, equation (5) was used to obtain the average coherence per subject, channel combination, and frequency band (α, β, γ). However, before applying equation (5) the distribution of the significant coherence features was evaluated using the Shapiro-Wilk test with a $p - value$ of 0.05 because the vector size was kept less than 50 samples. In the results obtained, it is concluded that the significant coherence data in the 3 frequency bands (α, β, γ), the 2 weight conditions (0.165 kg, 0.660 kg), the channels combination and the subjects follow a normal distribution with $p - value < 0.05$.

$$CS = \sum_{f=1}^{N} |X(f)|^2 \tag{5}$$

where the CS is the average coherence spectrum significant, f is the specific frequency significant, N is the number of significant coherences by the bands (α, β and γ) and X is the value of significant coherence. In this study, equation (5) was used for each independent frequency band, where to estimate coherence over the entire $(6 - 50)$ Hz spectrum, connectivity information was extracted from each frequency band and averaged.

2.3.4. Coherence analysis

Three different studies were performed to evaluate the coherence between EEG and EMG signals during the movement. In the first study, segmentation was performed in the frequency bands (α, β, and γ) where all the EMG channels were averaged for each EEG channel and each frequency band analyzed to determine in which frequency band significant differences exist between weights. For the second study, the coherence was evaluated for the combination of each EEG and EMG channel in the frequency band from 6 to 50 Hz to determine in which muscles there are significant differences between weights. Finally, coherence is evaluated for the 0.165 kg weight in each frequency band and in each muscle to determine under similar weight conditions which muscles are mostly in communication with each side of the motor cortex during these actions using information from EEG channels C_3 and C_4. For each study, the distribution of the data was assessed by applying the Shapiro-Wilk test with a $p - value$ of 0.05, where all data were found to have a high probability of following a normal distribution.

2.3.5. Statistical analysis

Statistical analysis was performed to identify significant differences between the connectivity of the two weights analyzed when the subject performs the manipulation of an object. For this, first, the Shapiro–Wilk test with a $p - value$ of 0.05 was applied to determine the normal distribution of the data and the Levene's test was applied to determine the homogeneity of variances between the data. Based on these results, the two-sample paired t-test was subsequently applied using Matlab because the data follow a normal distribution and equal variances. Significance levels for this test were set at 5% (**) and 10% (*). The alternative hypothesis is that one of the two weights present greater coherence than the other weight in the first two studies performed according to section 2.3.4, and the null hypothesis is the opposite. For the third study, the alternative hypothesis is that one of the two EEG channels (C_3 and C_4) presents greater coherence in each of the 3 frequency bands and the 5 muscles, and the null hypothesis is the opposite. For these purposes, the results are presented in bar figures with confidence intervals of 95% to evaluate the significant variation between subjects.

3. Results

Subjects perform an upper-limb movement that involves reaching, grasping, lifting, holding, and replacing an object of two different weights. Taking into

account that each subject performs the motor tasks at a different time, quantification of the time variation between subjects and between the tasks was obtained during the manipulation of the object.

Figure 4 shows the time variation between subjects for each task performed in the manipulation of the object (Reach, Grasp, Lift, Hold and Replace), the time variation of the whole task performed until the replacement of the object between subjects (blue for the weight of 0.165 kg and red for the weight of 0.660 kg), and the distribution of each task along with the movement of manipulating an object for the weight of 0.165 kg, where the red color represents the reaching task, blue represents the grasp, green represents the lift, yellow represents holding and magenta represents replacing the object. In Figure 4, the rest period corresponds to the first two seconds, and the time after the replace task corresponds to the release task. In the figure, the task with the longest duration within the manipulation of the object was the holding that lasts approximately 2 seconds, and the task with the shortest duration is the lifting that lasts approximately 0.30 seconds. On the other hand, the duration of the movement for the weight of 0.165 kg among the mean of the subjects corresponds to 6.88 ± 0.40 and for the weight of 0.660 kg corresponds to 7.55 ± 0.57.

In this article, data were taken for each trial until the subject performed the task of replacing the object. First, obtained results of the data integrity validation were presented. Next, coherence between EEG and EMG signals was shown in frequency bands and upper-limb muscles involved and finally, the study is presented to demonstrate which muscles are mostly in communication with the hemispheres of the motor cortex.

3.1. ERD/ERS quantification

Figure 5 presents the head maps for each analyzed weight using all trials and subjects spanning the frequency range $8-30$ Hz from $2-8$ seconds. According to the figure, the movement was distributed in all cortical cortex by the power decrease presented in the head map, where this effect could be presented by

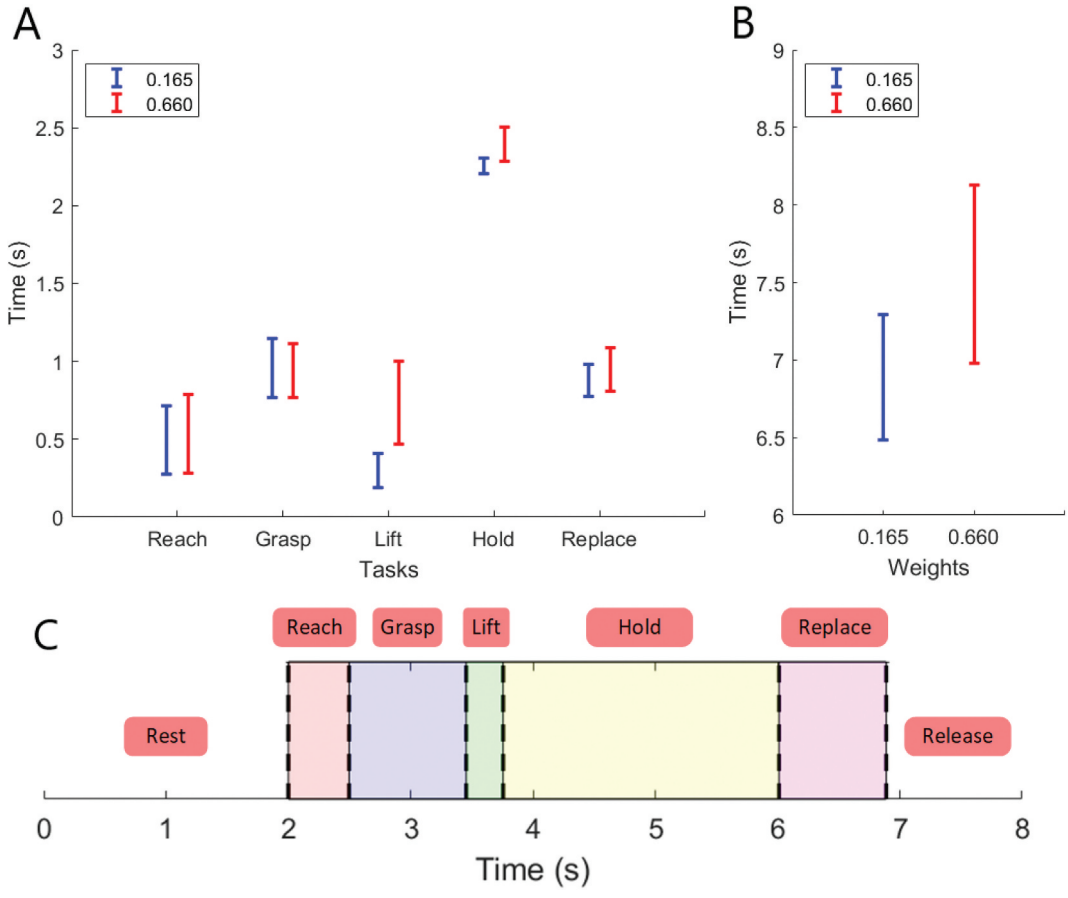


Figure 4. Duration of segments and trials. (A) Average of all subjects in each task involved in the movement. (B) Average of all subjects for the total duration of trials taken until the replacing the object. (C) Distribution of duration of the 0.165 kg weight for each task involved in the manipulation of the object.

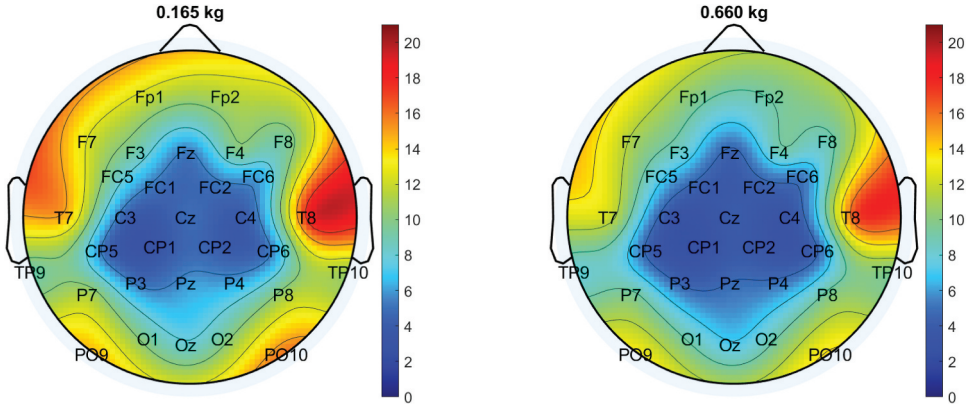


Figure 5. Power Spectral Density (PSD) for all EEG channels (32 channels) for both weights using all trials and subjects analyzed. PSD magnitude extends in the range $(0-20)$ of magnitude expressed in the head map for the execution movement $(2-8$ seconds) in the $8-30$ Hz frequency band.

the movement complexity. Finally, the EEG channels chosen for the coherence analysis were C_3, C_4 and C_z due to the main location in the cortical-motor cortex.

The ERD/ERS (relative power) was calculated to observe the change in power when the subject performs the movement with the two weights. The EEG time-frequency representations were calculated for C_3, C_4 and C_z channels in the $8-30Hz$ frequency band throughout the movement. Figure 6 shows time-frequency maps of ERD/ERS percentage values across all trials (55 trials per weight) and subjects (12 subjects) recorded for each weight. The figure is divided by α and β bands with a bar color between $(-50\%, 50\%)$. During the rest segment $(0-2$ seconds) there is no ERD/ERS before the vertical dotted line, as shown in Figure 6. After 2 seconds, a LED turns on and subjects start performing the movement; the ERD/ERS data reveals a significant ERD after the 2 seconds in all 3 channels according to the results of the bootstrap algorithm [25].

Figure 7 shows the ERD/ERS as a representation of the relative amplitude (%) versus the time (s) taken in the execution of the movement. The results are presented for the 3 EEG channels (C_3, C_z y C_4) segmenting the information α $(8-13$ Hz) and β $(14-30$ Hz) frequency bands, and presenting for the same channel the variation of the two weights, where the weight 0.165 kg is presented in blue color and the weight of 0.660 kg in red. However, for each frequency band, the mean and

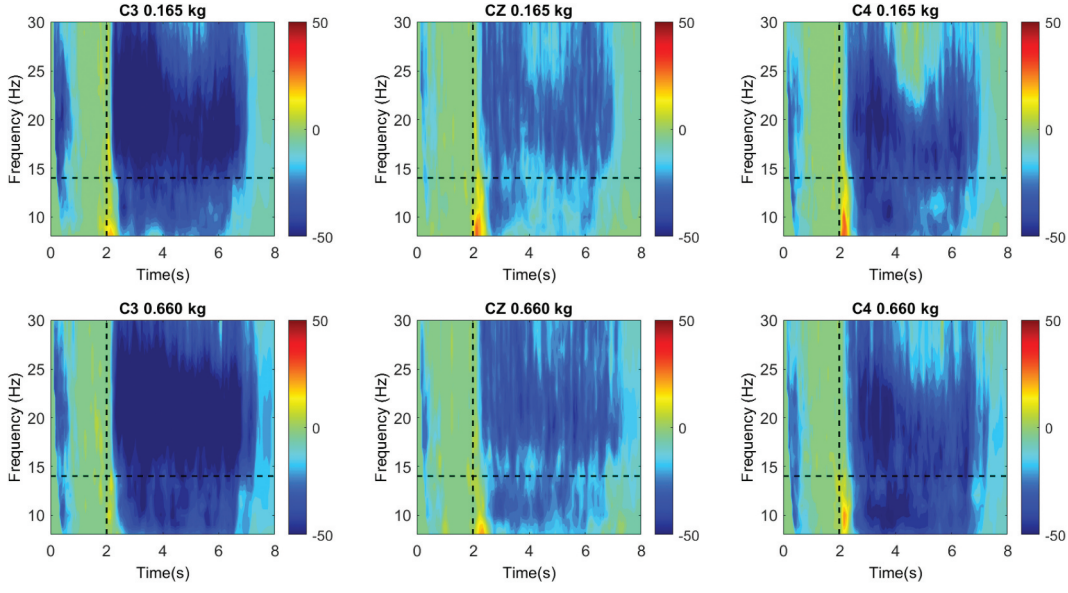


Figure 6. Time-frequency maps ERS/ERD for C_3, C_z and C_4 channels for two weights using all trials and subjects analyzed. The black vertical dotted line represents the time turning on the LED and subjects start to perform the movement. The black horizontal dotted line presents α and β bands separation.

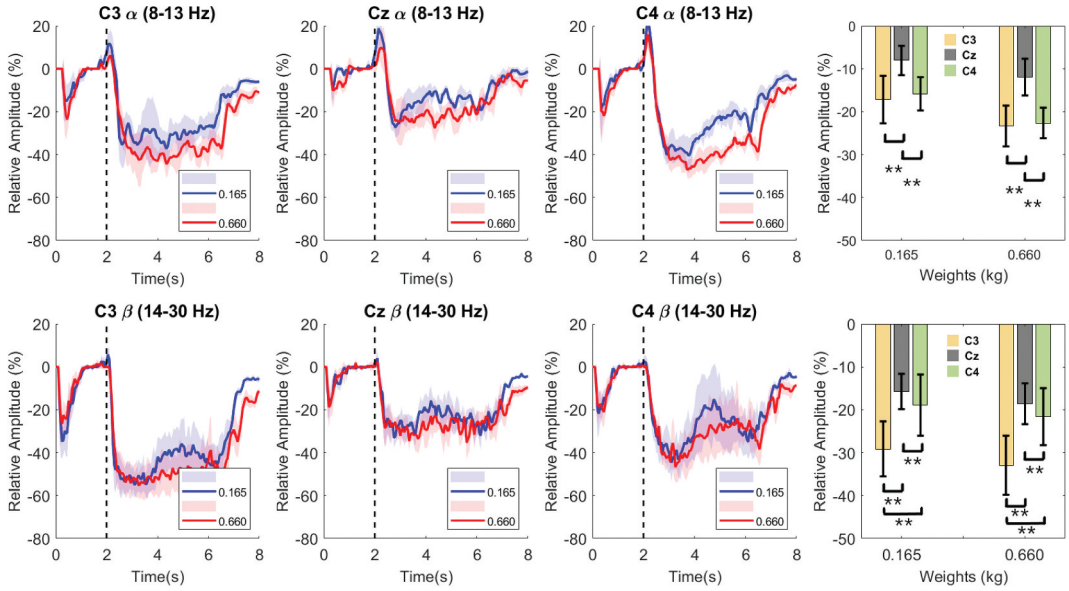


Figure 7. ERD/ERS value and its statistics. (A) Relative amplitude (%) for the 3 channels and two weights by segmenting the information into (α) and (β) frequency bands. The black line of vertical dots represents the moment when the LED is turned on and the subjects start to perform the movement. (B) Average and standard deviation of the relative amplitude of the 3 channels and the two weights for the two frequency bands. (**) Represents that there are significant differences ($p < 0.05$) between the channels.

standard deviation as shaded regions for the 3 channels in the two weights are presented, as shown in the bars of Figure 7.

To evaluate the contralateral behavior in the execution of the movement, a statistical analysis was performed to identify significant differences between the channels and between the weights. First, an analysis was performed using the Kolmogorov–Smirnov test (Samples $>$ 50 data) to confirm that the behavior of the ERD/ERS for each channel and weight have a high probability of having a normal distribution. Subsequently, Mann-Whitney U-test was applied with a value of $p = 0.05$ since the data did not present a normal distribution. According to the results, there are significant differences ($p < 0.05$) in the relative amplitude of the two weights in the 3 channels for the 2 frequency bands. Additionally, in α there are significant differences ($p < 0.05$) between channels C_3 and C_z, and C_4 and C_z for the two weights, and in β there are significant differences ($p < 0.05$) in the relative amplitude for the 3 channels (C_3 and C_z, C_3 and C_4, C_4 and C_z).

Considering the results presented in Figures 6 and 7, the 0.660 kg weight presents a greater decrease in power than the 0.165 kg weight and that in the C_3 channel there is a greater decrease in power compared to the C_z and C_4 channels, which could confirm the contralateral effect that occurs when movements are performed with the right hand. That is, the C_3 channel presents greater prominence than the C_4 channel when the movement was performed. In the literature, it is reported that when a user performs a right-hand motor task, an ERD is presented in the contralateral side (C_3 channel) [23].

3.2. Estimation and analysis of EMG amplitude

Changes in the electrical activity of the upper limb muscles (5 muscles) when the subject performs the movement was quantified. This muscle activity is expected to provide greater activation in the 0.660 kg weight than in the 0.165 kg weight. The RMS value of the EMG signal in the five muscles was calculated, as shown in Figure 8, where the 0.165 kg weight is presented by a blue line and the 0.660 kg weight is presented by a red line. The average value of muscle electrical activity for the 5 muscles of all recorded subjects is presented along with the standard deviation in shaded regions for the two weights, as shown in Figure 8. This allows us to observe how the change in muscle activation is for all subjects. However, according to the figure and the statistical analysis performed using the Mann-Whitney U-test with a significance value of 5%, the 0.660 kg weight presents greater muscle activation than the 0.165 kg weight evaluated for each muscle individually with a $p - value < 0.05$. As expected, there is no muscle activation before initiating the movement presented as a vertical dotted black line at second 2. All EMG channels (5 channels or muscles) were selected to evaluate the coherence between EEG and EMG signals in the analyzed movement.

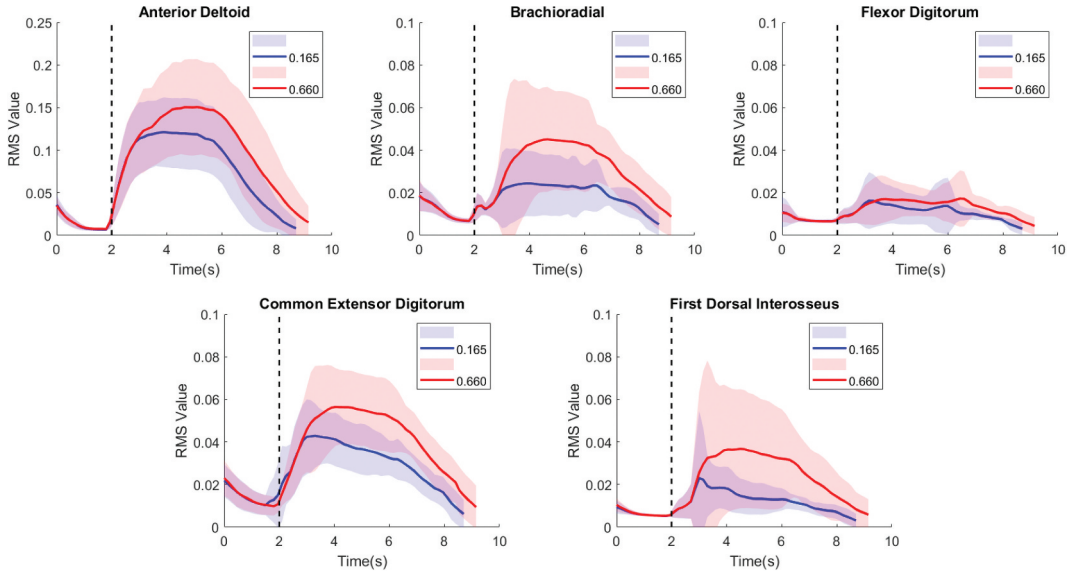


Figure 8. Root Mean Square (RMS) value for all subjects with the standard deviation presented in shaded regions for the two weights in the 5 muscles recorded. The black line of vertical dots represents the time at which the LED is turned on and at which the movement is initiated. The blue line presents the RMS value for the 0.165 kg weight and the red line for the 0.660 kg weight.

3.3. Coherence for frequency bands and muscles

Connectivity was quantified over the time interval from the start of the reach task (LED on) to the end of the replace task (see Figure 4). For this, the coherence spectrum for each EEG channel averaged over all EMG channels in the frequency bands (α, β, and γ) is shown in Figure 9(A). As an error bar, confidence intervals of 95% were calculated with respect to all subjects for each weight and EEG channel. In addition, the results of the statistical analysis using the two-sample paired t-test are presented, where (*) corresponds to the 10% significance level and (**) corresponds to the 5% significance level.

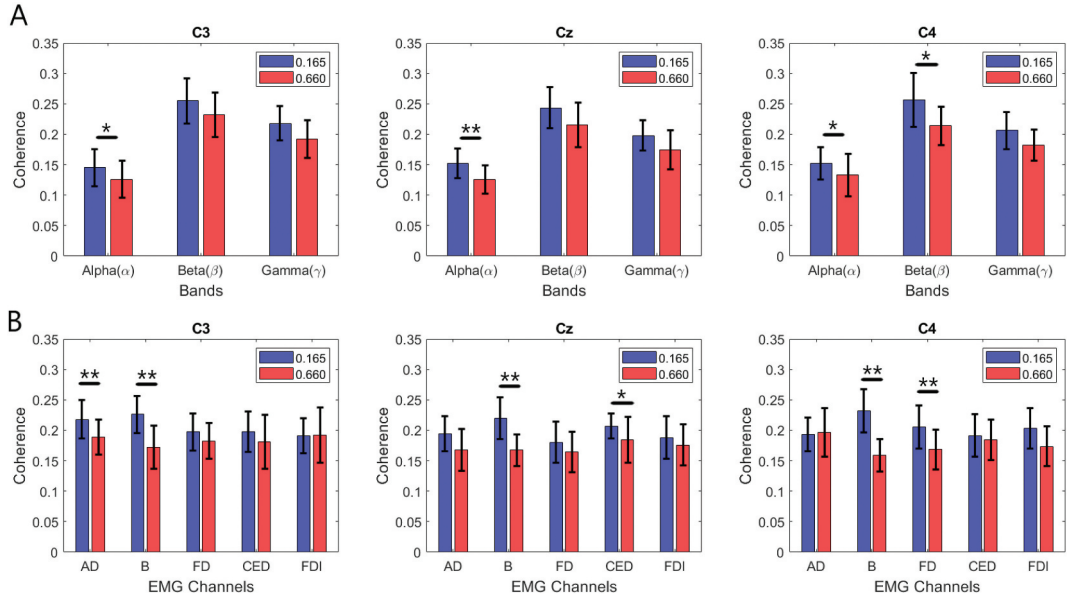


Figure 9. Coherence and statistical analysis for two weights quantified over the time interval from the start of the reach task (LED on) to the end of the replace task. (A) Coherence between each EEG channel (C_3, C_z, C_4) and all EMG channels in the 3 frequency bands for all subjects in the two weights. (B) Coherence between each EEG and EMG channel in the averaged of (α, β and γ) frequency bands for all subjects in the two weights. The confidence intervals of 95% regarding all subjects are presented. (*) corresponds to significant differences between weights at a significant level of 10% and (**) corresponds to a significant level of 5%.

The α band for the 3 EEG channels has a low coherence concerning other bands. On the other hand, the greatest coherence was found in the β band for the 3 channels, as shown in Figure 9(A). The γ band presents a coherence spectrum between the values of α and β band coherence. It is important to highlight that the coherence with the 0.165 kg weight is greater than the 0.660 kg weight for all bands at the 3 channels (C_3, C_z, C_4). Therefore, statistically significant differences were found for some cases as in the C_3 channel where there are significant differences $(p < 0.1, p = 0.08)$ in the coherence between weights for α band. Moreover, for the C_z channel there are significant differences of $(p < 0.05, p = 0.04)$ in the α band between weights and in the C_4 channel are significant differences in the α and β band with $p - values$ of $(p < 0.1, p = 0.09)$ and $(p < 0.1, p = 0.08)$, respectively.

However, the connectivity where muscle connections were varied regard each EEG channel (C_3, C_z, C_4) averaged over the 3 frequency bands (α, β, γ) is presented in Figure 9(B). This figure allows identifying how connectivity changes for each muscle, taking into account that each muscle presents a different electrical activation depending on the task performed in the manipulation of the object which is controlled by the Central Nervous System (CNS). The muscles are presented and labeled as Anterior Deltoid (AD), Brachioradial (B), Flexor Digitorum (FD), Common Extensor Digitorum (CED), and First Dorsal Interosseous (FDI). Connectivity was quantified over the time interval from the start of the reach task (LED on) to the end of the replace task (see Figure 4).

The coherence for the weight of 0.165 kg in the Brachioradialis muscle was higher than other muscles for all EEG channels. Coherence of the weight of 0.660 kg for Brachioradialis muscle is lower than others in the 3 EEG channels. Nevertheless, the coherence for the 0.660 kg weight was lower than the 0.165 kg weight for all muscles and EEG channels, except for the connectivity of C_3 with the FDI muscle where the coherence for both weights is approximately equal, and the connectivity between C_4 and the AD muscle where the coherence at the 0.660 kg weight is higher than the 0.165 kg weight. However, these results are not significant according to the statistical analysis. Significant differences were found as follows: For the C_3 channel, there were differences with a significant level of 5% for the AD and B muscle with a $p - value$ for both of $(p < 0.05, p = 0.04)$; for the C_z channel, there were significant differences with a 5% of significance considering the B muscle with a $p - value$ of $(p < 0.05, p = 0.03)$ and differences with a significant level of 10% for the CED muscle $(p < 0.1, p = 0.09)$. Finally, for C_4 channel, there were significant differences with a 5% of significance considering the B and FD muscles with $p - value$ $(p < 0.05, p = 0.00)$ and $(p < 0.05, p = 0.04)$, respectively. Brachioradial muscle presents significant differences between weights for the 3 EEG channels indicating that B muscle has an important change regarding the connectivity when the weight is changed.

3.4. Analysis of coherence according to hemispheric laterality

Figure 10 presents the muscles that are mostly in communication with each side of the motor cortex during the manipulation of an object using EEG–EMG connectivity quantified over the time interval from the start of the reach task (LED on) to the end of the replace task (see Figure 4). Initially, only the coherence calculated for the weight of 0.165 kg was used, where the variation of coherence for the 5 recorded muscles and the 3 frequency bands in channels C_3 and C_4 are presented. In the figure, C_3 channel is represented using yellow color and C_4 channel using green color. A statistical analysis using the two-sample paired t-test is presented in order to show whether there are significant differences in the coherence of the 2 EEG channels with respect to each frequency band and recorded muscles, where (*) corresponds to the 10% significance level and (**) corresponds to the 5% significance level.

The EEG channels (C_3 and C_4) present significant differences for the AD muscle in the α band at a significance level of 5% with $p - value$ $(p < 0.05, p = 0.03)$. For the B muscle these differences are found in the β band with 5% significance and in the γ band with 10% significance with $p - value$ of $(p < 0.05, p = 0.02)$ and $(p < 0.1, p = 0.09)$, respectively. However, for the FDI muscle the difference is in the α band at the 10% significance level with $p - value$ $(p < 0.05, p = 0.07)$. The muscles that do not present significant differences between the channels are the FD and CED in the 3 frequency bands. According to the results, there is a contralateral behavior in the α band of the AD muscle and in the γ band of the B muscle, and there is an ipsilateral behavior in the β band of the B muscle and in the α band of the FDI muscle.

4. Discussion

Obtained results of EEG–EMG coherence allow determine how functional corticomuscular connectivity occurs between brain cortical area and specific muscles when a reach-grasp-lift-hold and replace an object

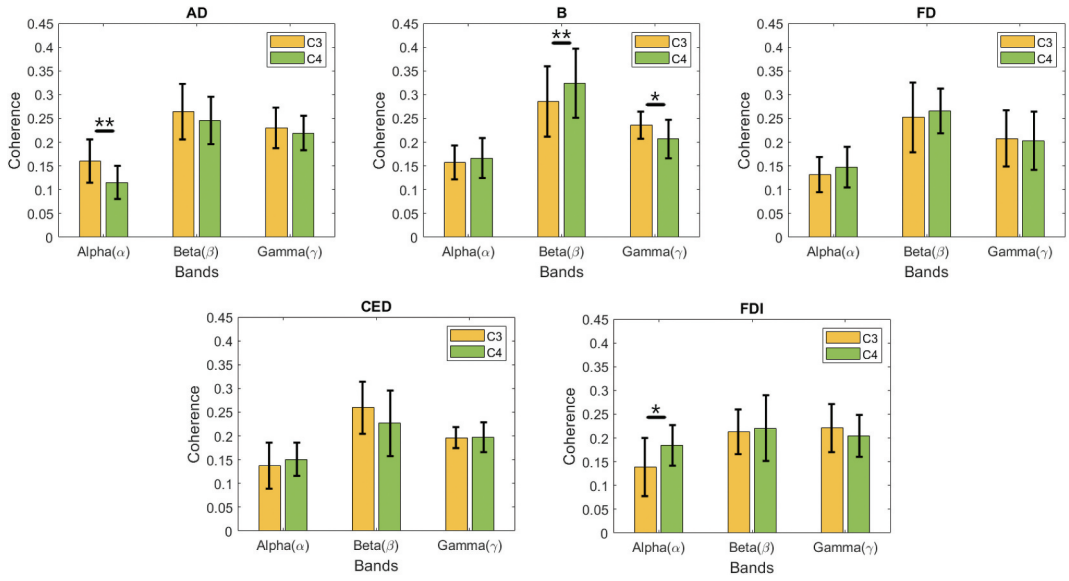


Figure 10. Coherence for the 0.165 kg weight in the EEG channels (C_3 and C_4), the 5 muscles and the 3 frequency bands (α, β and γ). The confidence intervals of 95% regarding all subjects are presented. Connectivity was quantified over the time interval from the start of the reach task (LED on) to the end of the replace task. (*) corresponds to significant differences between weights at a significant level of 10% and (**) corresponds to a significant level of 5%.

motor task was performed. The study aims to investigate how EEG and EMG connectivity changes during the manipulation of an object having a low and high weight (0.165 kg and 0.660 kg).

Cortical responses characterization of EEG data showed that when subjects perform the activity with the 0.660 kg weight presents a greater decrease in power than the 0.165 kg weight in the α and β frequency bands presenting a contralateral behavior. As mentioned above, the manipulation of an object involves different motor tasks such as reaching, grasping, lifting, holding, and replacing, which generate changes in brain signals according to the cellular excitability that can form each motor task. Being a complex task because multiple joints are involved in performing the movement, larger brain neural networks are needed in the processing of the information presented in the movement, so an ERD is present around the entire execution of the movement [23]. On the other hand, this type of task involves a number of different muscles, where muscle activation differs throughout the movement because each muscle is in charge of specific functions in the movement. As it is a combination of different tasks, the muscle synergies (muscle groups) are coactivated in a coordinated way through the CNS to perform a complex task. In other words, the CNS activates the muscle groups to generate a different motor command [34].

Significant differences in the movement of manipulation of an object with two different weights were found by quantifying the connectivity of EEG and EMG signals. Specifically, significant differences were found of; a) coherence of C_3 with Anterior Deltoid and Brachioradial muscles; b) coherence of C_z with Brachioradial muscle; c) coherence of C_4 with Flexor Digitorum and Brachioradial muscles. Furthermore, the greatest coherence was found in the β band for the 3 EEG channels, which is in agreement with the results of previous studies [10,12,13,35]. Those studies reported that subjects showed a peak in the coherence spectra between 15 – 30 Hz bandwidth during a hold task involving stable force production where the coherence between signals may be affected by the performance in the development of motor tasks. On the other hand, in future studies, we propose to evaluate how the connectivity is for each task involved in the manipulation of the object, in order to determine the delay times that occur in sending the information to execute each action involved in the movement [4].

Studies have been published in the literature that has determined contralaterality with respect to corticomuscular connectivity in a single muscle for upper limb movements. Effects have been found that relate to there being greater connectivity quantified using the coherence algorithm, in the contralateral hemisphere recorded by the channels (C_3 and C_4) establishing connectivity with only the Flexor-Digitorum-Superficialis (FDS) muscle when performing right and left-hand movements in healthy subjects [36]. However, that study quantifies corticomuscular connectivity in an upper limb movement when it reaches its final posture.

This study involves a multi-joint movement since the subjects perform a movement of control and precision with a synergy of synchronized movements to perform the action of manipulation of an object, which according to cortical topographies shows an ERD in the entire motor cortex of the brain.

Corticomuscular coherence has been shown to be task-dependent, that it reflects attention and precision, compliance of the gripped objects, displacement, magnitude of the force, and learning processes [2]. Thus, effects such as motor adaptation or neuronal noise sources may be taking place in the modulation of connectivity during the manipulation of the object involving two weights [21,37]. Furthermore, corticomuscular coherence has widely been considered a motor mechanism reflecting efferent oscillatory activity descending from the primary motor cortex via corticospinal pathways, but it has been reported works that suggest that ascending sensory activity also contributes to the coupling between sensorimotor brain regions and muscle [38]. Such phenomenon could be taking part in obtained significant differences of coherence during reach-to-grasp movement for two different weights.

The results from the coherence analysis show that there are significant differences in EEG–EMG connectivity in the manipulation of an object of two different weights (low and high) mainly in the α and β band, and in the AD, B, FD and CED muscles in different EEG channels. These differences are centered on the 0.165 kg weight presents higher coherence than the other weight. Therefore, by applying these data in a multimodal acquisition system, a high classification could be obtained in the identification of motor tasks, not only when the upper limb has already reached its final position but also in the identification of different tasks that are involved in a given action such as the manipulation of an object.

The findings of this study could be of great importance in the development of Hybrid Brain-Computer Interface (hBCI) systems because currently, the implementation of a BCI system based exclusively on EEG signals generates low accuracy in pattern decoding, due to the fact that it only generates one control command [4,18]. However, hBCIs have 3 different ways to be implemented characterized in multiple biosignals such as EEG–EMG, paradigms and multisensory stimuli [39] which allows increasing the reliability of the system in terms of the percentage of command classification in specific tasks and the generalization of applications in fields such as Neuro-Rehabilitation [17, 18, 39, 40]. Mainly, the advantages found encompass (1) higher BCI classification accuracy, (2) higher number of brain commands for control application, and (3) shorter brain command detection time [17,41]. Therefore, these provide a benefit and a clear advantage over any single brain signal acquisition modality.

In this work, EMG-EEG coherence is higher in the beta frequency band were demonstrated, individualized differences in coherence have been found according to each muscle involved during the reaching and grasping movement, showing that the brachioradialis muscle is the most involved in the connectivity due to the significant differences found in the EEG channels. Additionally, it has been demonstrated how the EMG-EEG coherence could change depending on the force exerted to grasp an object of different weight, and it has been determined which muscles are mostly in communication with each side of the cerebral hemispheres.

Finally, the results presented in this study allow us to conclude that coherence is significantly higher at a weight of 0.165 kg than at a weight of 0.660 kg for the α band in the C_3, C_z and in α and β for the C_4 channel. In addition, these differences are found in the C_3-AD, C_3-B, C_z-B, C_z-CED, C_4-B and C_4-FD channel combinations. These results are of great importance in rehabilitation engineering applications because corticomuscular connectivity can be used as a descriptor to improve the classification rates, usability, and control of prostheses based on BCI systems. In this case, the application of prostheses for weightlifting identification. However, as future studies, we propose to evaluate other methods such as Granger Causality to establish connectivity and delay times between EEG and EMG information when performing other types of movements involved in activities of daily living. As well as using corticomuscular connectivity as a rehabilitation method for people with disabilities, for implementing computational methods based on connectivity to improve identification rates using techniques such as filter banks.

Acknowledgment

The authors would like to thank the Antonio Nariño University, for the support of the development in this work.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by Antonio Nariño University (UAN) under the grant N.2021020 'Model based on multimodal EEG–EMG information to improve motion intention decoding for the control of a BCI system'.

ORCID

Cristian D. Guerrero-Mendez http://orcid.org/0000-0003-1019-9423
Andres F. Ruiz-Olaya http://orcid.org/0000-0002-5883-5786

References

[1] Conway B, Halliday D, Farmer S, et al. Synchronization between motor cortex and spinal motoneuronal pool during the performance of a maintained motor task in man. J Physiol. 1995;489(3):917–924.
[2] Andrykiewicz A, Patino L, Naranjo JR, et al. Corticomuscular synchronization with small and large dynamic force output. BMC Neurosci. 2007;8(1):1–12.
[3] Rizzo R, Zhang X, Wang JW, et al. Network physiology of cortico–muscular interactions. Front Physiol. 2020;11. DOI:10.3389/fphys.2020.558070
[4] Kim B, Kim L, Kim Y-H, et al. Cross-association analysis of EEG and EMG signals according to movement intention state. Cognit Syst Res. 2017;44:1–9.
[5] Zhuang KZ, Lebedev MA, Nicolelis MA. Joint cross-correlation analysis reveals complex, time-dependent functional relationship between cortical neurons and arm electromyograms. J Neurophysiol. 2014;112(11):2865–2887.
[6] Liu J, Sheng Y, Liu H. Corticomuscular coherence and its applications: a review. Front Hum Neurosci. 2019;13:100.
[7] Mima T, Hallett M. Electroencephalographic analysis of cortico-muscular coherence: reference effect, volume conduction and generator mechanism. Clin Neurophysiol. 1999;110(11):1892–1899.
[8] Rossiter HE, Eaves C, Davis E, et al. Changes in the location of cortico-muscular coherence following stroke. NeuroImage Clin. 2013;2:50–55.
[9] Sburlea AI, Müller-Putz GR. Exploring representations of human grasping in neural, muscle and kinematic signals. Sci Rep. 2018;8(1):1–14.
[10] Boonstra TW, van Wijk BC, Praamstra P, et al. Corticomuscular and bilateral EMG coherence reflect distinct aspects of neural synchronization. Neurosci Lett. 2009;463(1):17–21.
[11] Lóopez-Larraz E, Birbaumer N, and Ramos-Murguialday A. 2018. A hybrid EEG- EMG BMI improves the detection of movement intention in cortical stroke patients with complete hand paralysis. In *2018 40th annual international conference of the IEEE engineering in medicine and biology society (EMBC)* Honolulu, HI, USA (pp. 2000–2003).
[12] Gao L, Wu H, Cheng W, et al. Enhanced descending corticomuscular coupling during hand grip with static force compared with enhancing force. Clin EEG Neurosci. 2020;52(6):436–443.
[13] Kilner J, Baker S, Salenius S, et al. Task- dependent modulation of 15-30 Hz coherence between rectified EMGs from human hand and forearm muscles. J Physiol. 1999;516(2):559–570.
[14] Borhani S, Abiri R, Jiang Y, et al. Brain connectivity evaluation during selective attention using EEG-based brain-computer interface. Brain-Comput Interfaces. 2019;6(1–2):25–35.
[15] Guerrero-Mendez CD, Blanco-Diaz CF, and Ruiz-Olaya AF. 2021. Identification of motor imagery tasks using power-based connectivity descriptors from EEG signals. In *2021 XXIII symposium on image, signal processing and artificial vision (STSIVA)* Popayán, Colombia (pp. 1–6).
[16] Li X, Samuel OW, Zhang X, et al. A motion- classification strategy based on sEMG-EEG signal combination for upper-limb amputees. J Neuroeng Rehabil. 2017;14(1):1–13.
[17] Hong K-S, and Khan MJ. Hybrid brain–computer interface techniques for improved classification accuracy and increased number of commands: a review. Front Neurorobot. 2017 July;11 35 .
[18] Ma T, Li H, Deng L, et al. The hybrid BCI system for movement control by combining motor imagery and moving onset visual evoked potential. J Neural Eng. 2017 February;14(2):026015.
[19] Betti S, Zani G, Guerra S, et al. Reach-to-grasp movements: a multimodal techniques study. Front Psychol. 2018;9:990.
[20] Batzianoulis I, El-Khoury S, Pirondini E, et al. EMG-based decoding of grasp gestures in reaching-to-grasping motions. Rob Auton Syst. 2017;91:59–70.
[21] Mendez-Balbuena I, Huethe F, Schulte-Mönting J, et al. Corticomuscular coherence reflects interindividual differences in the state of the corticomuscular network during low-level static and dynamic forces. Cereb Cortex. 2012;22(3):628–638.
[22] Steriade M, Llinás RR. The functional states of the thalamus and the associated neuronal interplay. Physiol Rev. 1988;68(3):649–742.
[23] Pfurtscheller G, Da Silva FL. Event-related EEG/MEG synchronization and desynchronization: basic principles. Clin Neurophysiol. 1999;110(11):1842–1857.
[24] Luciw MD, Jarocka E, Edin BB. Multi-channel EEG recordings during 3,936 grasp and lift trials with varying weight and friction. Sci Data. 2014;1(1):1–11.
[25] Graimann B, Huggins JE, Levine SP, et al. Visualization of significant ERD/ERS patterns in multichannel EEG and ECoG data. Clin Neurophysiol. 2002;113(1):43–47.
[26] Merletti R, Botter A, Troiano A, et al. Technology and instrumentation for detection and conditioning of the surface electromyographic signal: state of the art. Clin Biomech. 2009;24(2):122–134.
[27] Renshaw D, Bice MR, Cassidy C, et al. A comparison of three computer-based methods used to determine EMG signal amplitude. Int J Exercise Sci. 2010;3(1):43.
[28] Dimigen O. Optimizing the ICA-based removal of ocular EEG artifacts from free viewing experiments. NeuroImage. 2020;207:116117.
[29] Krauledat M, Dornhege G, Blankertz B, et al. Robustifying EEG data analysis by removing outliers. Chaos Complexity Letters. 2007;2(3):259–274.

[30] Xi X, Ma C, Yuan C, et al. Enhanced EEG-EMG coherence analysis based on hand movements. Biomed Signal Process Control. 2020;56:101727.
[31] Hashimoto Y, Ushiba J, Kimura A, et al. Correlation between EEG-EMG coherence during isometric contraction and its imaginary execution. Acta Neuro biol Exp (Wars). 2010;70(1):76–85.
[32] Bortel R, Sovka P. EEG-EMG coherence enhancement. Signal Process. 2006;86(7):1737–1751.
[33] Rosenberg J, Amjad A, Breeze P, et al. The Fourier approach to the identification of functional coupling between neuronal spike trains. Prog Biophys Mol Biol. 1989;53(1):1–31.
[34] Torricelli D, Barroso F, and Coscia M, et al. 2015 October. Muscle synergies in clinical practice: theoretical and practical implications Pons, J, Raya, R, and González, J. In: Biosystems & biorobotics. 251–272. Switzerland: Springer International Publishing.
[35] Kristeva R, Patino L, Omlor W. Beta-range cortical motor spectral power and corticomuscular coherence as a mechanism for effective corticospinal interaction during steady-state motor output. Neuroimage. 2007;36 (3):785–792.
[36] Chowdhury A, Raza H, Meena YK, et al. An EEG- EMG correlation-based brain-computer interface for hand orthosis supported neuro-rehabilitation. J Neurosci Methods. 2019;312:1–11.
[37] Trenado C, Mendez-Balbuena I, Manjarrez E, et al. Enhanced corticomuscular coherence by external stochastic noise. Front Hum Neurosci. 2014;8:325.
[38] Beck MM, Spedden ME, Lundbye-Jensen J. Reorganization of functional and directed corticomuscular connectivity during precision grip from childhood to adulthood. Sci Rep. 2021;11(1):1. November.
[39] Li Z, Zhang S, Pan J. Advances in hybrid brain-computer interfaces: principles, design, and applications. Comput Intell Neurosci. 2019 October;2019:1–9.
[40] Amiri S, Fazel-Rezai R, Asadpour V. A review of hybrid brain-computer interface systems. Adv Hum Comput Interact. 2013;2013:1–8.
[41] Banville H, Falk T. Recent advances and open challenges in hybrid brain- computer interfacing: a technological review of non-invasive human research. Brain-Comput Interfaces. 2016;3(1):9–46.